

2. Continuous random variables		
2.1	The concept of a continuous random variable.	
2.2	The probability density function and the cumulative distribution function for a continuous random variable.	Use of the probability density function $f(x)$, where $P(a < X \leq b) = \int_a^b f(x) \, dx .$ Use of the cumulative distribution function $F(x_0) = P(X \leq x_0) = \int_{-\infty}^{x_0} f(x) \, dx .$ The formulae used in defining $f(x)$ will be restricted to simple polynomials which may be expressed piecewise.
2.3	Relationship between density and distribution functions.	$f(x) = \frac{dF(x)}{dx} .$
2.4	Mean and variance of continuous random variables.	
2.5	Mode, median and quartiles of continuous random variables.	